

A Virtual Farm Tour for Public Education about Dairy Industry

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Figure 1: A captured scene of the developed virtual dairy farm simulation.

ABSTRACT

In this paper, we present an immersive virtual farm simulation developed to provide realistic on-farm experiences to the public. Users could visit the virtual farm, walk through various sites where dairy cows are raised, and learn how the dairy products are produced through the virtual experience. Public users' responses about the virtual experience were collected in various measures, e.g., user experience and learning efficacy, via showcases at local libraries. We present preliminary results regarding the potential of the simulation as an effective agricultural education, and discuss our future plans.

Index Terms: Applied computing—Education—Interactive learning environments; Applied computing—Computers in other domains—Agriculture; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

The public is becoming increasingly interested in investigating the origin of their food and how it is produced, with concerns about environmental impact, local sourcing, and ethics [2, 7]. However, the average North American supermarket shopper is detached from how the food they consume is actually produced. There is a knowledge-trust gap between consumers in Canada and the agriculture industry,

and the information being distributed by the industry is not reaching the public [9]. For example, shoppers are used to seeing their containers of milk and bricks of cheese on the supermarket shelves, but may have never seen a cow or thought about how the livestock that produces these goods are raised on farms outside their city. Others have reported that journalism can just as easily influence one's food consumption behaviour [8], and social media is becoming increasingly powerful in affecting public opinion based on the viewer's connection to the source [10]. Virtual Reality (VR) represents a promising technology to provide an accessible and interactive experience learning about the dairy industry, as it has been reported to be an effective education tool [3]. While many 2D farming video games, educational programs, and even some 3D farming VR gaming options are available [4, 5], those experiences often only offer a limited sense of how a niche industry or boutique farm operates.

VR has the potential to make sure agriculture information reaches the public in a meaningful way. Anastasiou et al. reviewed applications of VR and extended reality within the agriculture industry and found 55 studies, half of which are educational applications [1]. Sutherland et al. recommended using social media, school curriculum, and senior's programs to quickly reach the public [9]. As a supplemental or alternative to those conventional, VR as a communication tool has the potential to create a strong connection with the public. An immersive environment of a verifiable situation can provide the public with a meaningful experience to form their opinions.

In this paper, we introduce our project where we developed a VR simulation application to provide realistic on-farm experiences to the public. Using the developed application through wearable VR headsets, the public could walk through various sites where dairy cows are raised. We focus on realizing an interactive, scientifically accurate on-location experience that teaches the public about agriculture through a visit to a virtual farm for the day. The program was

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showcased at the University of Calgary's VR studio and the Calgary Public Library to examine the potential of the VR experience as a public education tool, while collecting various data from the public, e.g., user experience, learning efficacy, and thoughts/perception about agricultural production/industry. Here we present brief, yet positive preliminary results of the users' responses.

2 VIRTUAL DAIRY FARM SIMULATION

In this section, we describe the details of how we designed and developed our virtual dairy farm simulation.

2.1 Local Farm Visits and Consultation

To design an accurate and realistic dairy farm simulation, we consulted with both local farmers and international researchers. For instance, we talked with Dr. Maxime Delsart (Veterinary Medicine, École Nationale Vétérinaire d'Alfort) to learn from his experience in developing a 360-degree video-based immersive farm simulation. The video experience that they developed was an educational tool for veterinary students to learn about biosecurity rules on pig farms. The research team also visited local dairy farms to collect photos, videos, and information about farm operations in Alberta, Canada¹. To capture the correct and realistic representation of the real dairy farm and industry, we completed a series of prototype testing and ground-trusting with farmers, veterinarians, dairy industry representatives, and animal welfare experts.

2.2 Virtual Farm Design and Development

Farm Layout and Experience Design Using our collected data, we iterated upon our design until we completed a model with five main sections: feeding, veterinary care, milking, calving, and cross-sectional boundaries (Figure 2). Our model simulated a continuous indoor environment so that users could navigate and learn freely. We included four interactive virtual human agents to provide information about dairy production to the user. These agents included a nutritionist, a veterinarian, and two farmers. To provide an immersive experience, we integrated nonverbal gestures while the agents verbally described their role on the farm to the user.

Users start their dairy farm experience at the cross-section boundary, and can then move freely around the entire dairy barn. In the cross-section boundary, the nutritionist agent appears and talks about the dairy cows' feed and nutritional requirements. The area also contains information stalls that the user can interact with to watch or read educational material (e.g., text, images, and videos) prepared by experts and farmers about public concerns, animal welfare, and technologies used to collect data in dairy farms. In the feeding and veterinary areas, users can watch dairy cows rest and eat. Here, a farmer and a veterinary agent explain their jobs and what it takes to keep the environment and the cows healthy (Figure 3(a)). In the milking section, users can see cows being prepared for milking. The cows stand in a rotating milk parlor next to a cooling tank that stores the collected milk (Figure 3(b)). The second farmer agent is standing next to the parlor and describes the milking process, as well as his role in the process. The next area is the calving area, where newborn calves and mother cows can be observed. Users can call one of the calves to the fence to give water to them.

Simulation Development We used commercial off-the-shelf virtual models to model some of the cows, but we also hired a graphic designer to create 3D models and animation schemes in order to improve the realism of the experience. In addition, we included recurrent background sounds recorded on a real farm to enhance the realism. Unity Gaming Engine was used to create our digital farm model, and we used the OpenXR plugin to ensure that the simulation

¹Breakfast on the Dairy Farm: <https://www.familyfuncanada.com/calgary/breakfast-dairy-farm/> (Accessed 2023-01-21)

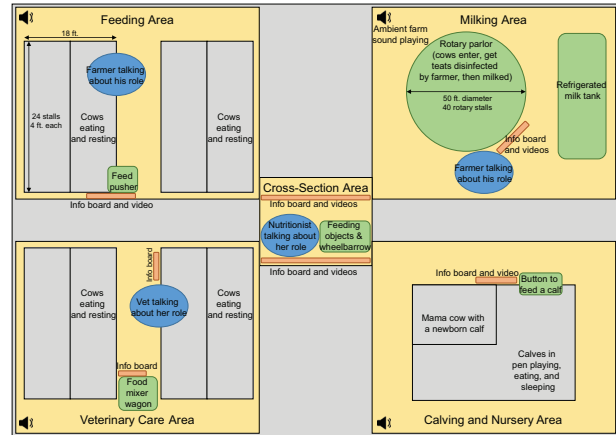


Figure 2: A layout of our virtual farm design, which consists of five areas: feeding, milking, veterinary, calving, and cross-section areas.

was compatible with multiple VR headsets for scalability. The final application was deployed on Oculus Quest 2.

For the user to interact with the virtual environment, we allowed them to use only the right-hand controller to streamline their experience. We used the XR Interaction Toolkit² for camera control, which allows the user to rotate and teleport around the virtual space. We restricted access to certain parts of the virtual space to avoid collisions between the user and the various models we used. For the interaction of calling a calf in the calving area, collision detection at the fences will trigger a calf to walk toward the user for an immersive experience. We also aimed to create realistic human models and animations to provide a realistic experience for the general public. We used the Microsoft Rocketbox library [6], which came with the animation clips that we used. Additionally, we recorded human voices for character audio and utilized Salsa LipSync Suite³ to make realistic lip-sync.

Since our target users may not have any prior experience with VR devices, we created a tutorial to show the users how to use the controller to navigate and interact with the virtual environment. Users are free to skip this tutorial if they choose.

3 SHOWCASES AND PRELIMINARY RESULTS

To validate the potential of our virtual farm simulation for public education about dairy production industry, "VR Dairy Farm" exhibitions were open to the public in Taylor Family Digital Library (University of Calgary) and Calgary Public Library in November 2022⁴. We hired students, as facilitators, to monitor the showcase operation and guide the public users, who participated in the virtual farm experience. All the participants were visitors to the two libraries, who were interested in the experience.

The participants were given information sheets about the simulation and potential risks accompanied by using VR headsets in advance. The facilitator helped them adjust the headset until they could see the virtual scene clearly and hold the right-hand controller. Participants were required to sit on a swivel chair during the experience to avoid potential experience of cybersickness and any risk

²Unity XR Interaction Toolkit: <https://docs.unity3d.com/Packages/com.unity.xr.interaction.toolkit@2.2/manual/index.html> (Accessed 2023-01-21)

³Salsa LipSync Suite: <https://crazyminnowstudio.com/docs/salsa-lip-sync/> (Accessed 2023-01-21)

⁴Experience a Dairy Farm in Virtual Reality: <https://www.simpsoncentre.ca/events/VRdairyfarm/> (Accessed 2023-01-21)



(a) The feeding area with a virtual farmer.



(b) The milking area with a rotary parlor.

Figure 3: Sample scenes captured in the virtual dairy farm simulation.

of physical injury. The facilitator suggested they go through the tutorial, but it is optional. Participants then started their experience in the cross-section and freely explored the barn as long as they want. After completing the experience, they could choose to fill out the questionnaires voluntarily.

Fifty-nine users (age $M = 32.32$ years, $SD = 11.22$ years, range: 17–60 years) voluntarily participated in the study by submitting their responses to the questionnaires after the experience. Among many other measures, here we only report several as preliminary results.

First, we wanted to see how authentic, useful, and recommendable the participants perceived the virtual experience was. We prepared a one-question survey for each item using 5-point Likert scale (1: Strongly Disagree, 5: Strongly Agree) that directly asked the participants' perception.

- **Provision of Authentic Information:** "The things I saw in the VR experience seemed trustworthy and unbiased."
- **Usefulness as an Education Tool:** "This VR experience is a useful platform for public agricultural education."
- **Willingness to Recommend:** "I would recommend this VR experience to my friends and family."

Overall, we found that the participants' responses to these questions were generally positive after the experience (Figure 4). The participants perceived that the information provided in the virtual farm experience is trustworthy and unbiased ($Mdn = 4$, $M = 3.72$, $SD = 1.18$). More importantly, they perceived it as a useful/helpful tool for public agricultural education ($Mdn = 4$, $M = 4.33$, $SD = 0.69$). Most participants indicated that they would recommend this experience to their acquaintances ($Mdn = 4$, $M = 4.39$, $SD = 0.70$).

Additionally, we asked another question, "What words describe your VR experience?" where multiple choices were available. A majority of the participants positively reported that the experience was educational ($N = 31$), fun ($N = 28$), and exciting ($N = 25$) (Figure 5). The number of participants who chose negative words, e.g., confus-

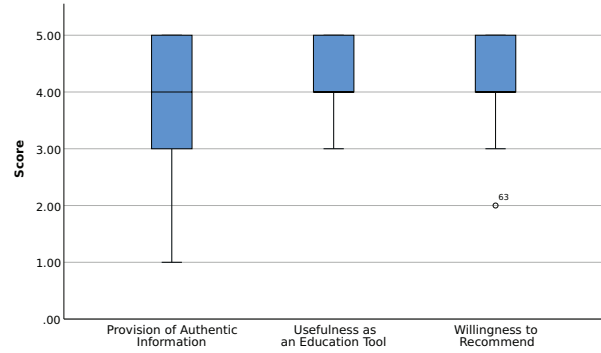


Figure 4: Results of the participant-perceived information authenticity, usefulness, and willingness to recommend.

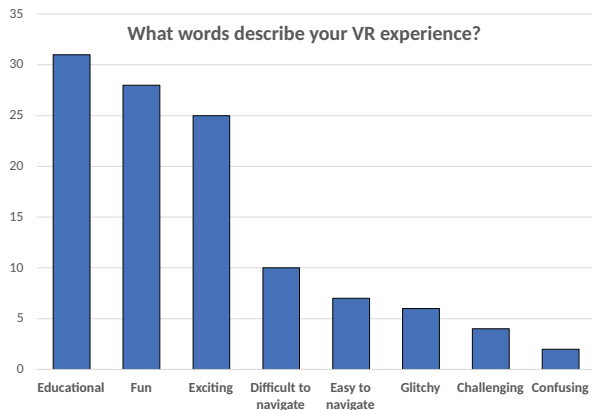


Figure 5: Results of showcase participants' responses for a question, "What words describe your VR experience?" The number of participants who selected the word is shown.

ing, challenging, and glitchy was fewer than 10. This indicates that the developed virtual farm simulation was perceived as a fun and exciting, yet effective learning tool for public education about dairy production.

4 SUMMARY AND FUTURE WORK

It is important to bring accurate information about agricultural production, so that they can understand how farmers produce the food we eat and develop trust in the industry. Bringing transparency to the agri-food system through immersive VR experiences can build public trust in food and farmers, address hesitation or misunderstanding about production practices, such as the use of new technologies, and help consumers make more informed choices at the grocery store.

In this paper, we presented an immersive VR-based dairy farm simulation for public users (e.g., dairy product consumers) to experience and learn about how the dairy products are actually produced in healthy farm environments. The preliminary results showed that the developed system was perceived as an effective and useful learning tool to provide authentic information to the consumers. We plan to investigate the effects of our virtual dairy farm experience and its realism on learning efficacy, including knowledge acquisition and retention through a controlled study.

This project will continue extending the features of the VR experience, e.g., more interactive activities with virtual entities in the simulation for better learning outcomes and motivation, and multi-user communication for the connection of farmers and consumers in

the virtual space. We also plan to create different virtual simulations beyond the scope of dairy production, to cover various types of agri-food industry, e.g., pork, poultry, and aquaculture.

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